



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Charge and Current — $Q = It$

One equation, one trap. The trap is the units.

Key concept

CURRENT IS A FLOW OF CHARGE

Current is not “electricity”. It is the **rate at which charge flows** — how much charge goes past a point each second.

$$\text{charge flow} = \text{current} \times \text{time} \quad Q = It$$

Q = charge flow, in **coulombs (C)** · I = current, in **amps (A)** · t = time, in **seconds (s)**

Rearranged: $I = \frac{Q}{t}$ and $t = \frac{Q}{I}$

The trap, and it is the one that costs marks. t must be in **seconds**. Real questions give you *minutes*, or hours. Convert **first**, and write the converted number down so a marker can see it.

1 minute = 60 s · **2 minutes** = 120 s · **1 hour** = 3600 s

Worked example

A current of 0.25 A flows for **3 minutes**. Calculate the charge that flows.

Convert first. $t = 3 \times 60 = 180$ s $Q = It = 0.25 \times 180 = \mathbf{45\ C}$

Using $t = 3$ would give 0.75 C — sixty times too small, and no method mark for a conversion you never wrote down.

1. State what is meant by *electric current*. [1]

The rate of flow of electrical charge.

2. Write down the equation that links charge flow, current and time. [1]

charge flow = current $\times$ time ($Q = It$)

3. A doorbell draws a current of 0.85 A for 40 seconds. Calculate the charge flow. **Give your answer to 2 significant figures.** [3]

$Q = It = 0.85 \times 40 = 34\ C = \mathbf{34\ C}$ (2 s.f.)

4. A current of 1.75 A flows in a heater element for 80 seconds. Calculate the charge flow. [2]

$Q = It = 1.75 \times 80 = \mathbf{140\ C}$

5. There is a current of 0.045 A in an LED for 240 seconds. Calculate the charge flow. [2]

$Q = It = 0.045 \times 240 = \mathbf{10.8\ C}$

6. A charge of 624 C flows through a motor in 130 s. Calculate the current. [2]

$I = Q/t = 624/130 = \mathbf{4.8\ A}$

7. A phone charger supplies a current of 0.90 A for **2 minutes**. Calculate the charge that flows. [3]

$$t = 2 \times 60 = 120 \text{ s.} \quad Q = 0.90 \times 120 = \mathbf{108 \text{ C}}$$

8. A charge of 1500 C flows through a lamp in **5 minutes**. Calculate the current. [3]

$$t = 300 \text{ s.} \quad I = 1500/300 = \mathbf{5.0 \text{ A}}$$

9. A torch bulb carries a current of 0.24 A. How long must it be switched on for 72 C of charge to flow? Give your answer **in minutes**. [3]

$$t = Q/I = 72/0.24 = 300 \text{ s} = \mathbf{5 \text{ minutes.}}$$
 Convert back — the question asked for minutes.

10. A student says: “A *bigger current always means more charge has flowed.*” Explain why this is not necessarily true. [2]

*Charge depends on current **and** time. A small current flowing for a long time can move more charge than a large current flowing for a moment.*

11. A single closed loop carries a current of 2.0 A. Explain why the same charge flows past every point in that loop in one minute. [2]

The current is the same at every point in a single closed loop, and the time is the same, so $Q = It$ gives the same value everywhere. Charge is not used up as it goes round.

Extension — how big is a coulomb, really?

Adapted from College Physics by OpenStax College (cnx.org), via Lumen Learning “20.1 Current”. Used under CC BY 4.0.

Background (not on the AQA specification — you will not be asked for this)

You have spent this sheet calculating charges of 34 C, 108 C, 140 C. Reasonable-sounding numbers. But **how much charge is one coulomb?**

A single electron carries a charge of

$$e = 1.6 \times 10^{-19} \text{ C}$$

which is a fantastically small amount. So the number of electrons needed to make up a charge Q is

$$\text{number of electrons} = \frac{Q}{e}$$

It is here because a coulomb is otherwise just a word — and because the answer is genuinely surprising.

Worked example

The doorbell in question 3 moved 34 C of charge. How many electrons is that?

$$\text{number} = \frac{Q}{e} = \frac{34}{1.6 \times 10^{-19}} = \mathbf{2.1 \times 10^{20} \text{ electrons}}$$

That is 210,000,000,000,000,000 electrons — for one press of a doorbell.

12. A torch bulb carries a current of 0.30 A for 60 seconds. Calculate the charge that flows, and then the number of electrons. [3]

$$Q = It = 0.30 \times 60 = 18 \text{ C.} \quad \text{number} = 18/(1.6 \times 10^{-19}) = \mathbf{1.1 \times 10^{20} \text{ electrons.}}$$

13. A current of 5.0 A flows in a kettle element for 2 minutes. Calculate the number of electrons that pass through it. [3]

$$t = 120 \text{ s.} \quad Q = 5.0 \times 120 = 600 \text{ C.} \quad \text{number} = 600 / (1.6 \times 10^{-19}) = \mathbf{3.8 \times 10^{21} \text{ electrons.}}$$

One more surprise, and it is worth knowing. Those electrons are not racing along the wire. In a copper wire carrying a normal current they *drift* at well under a millimetre per second — slower than you can walk, slower than a snail.

So why does a lamp light the *instant* you close the switch?

14. Explain why a lamp lights immediately when the switch is closed, even though the electrons themselves move extremely slowly. [3]

The wire is already full of free electrons along its whole length, including the ones inside the lamp. Closing the switch sets up the effect of the potential difference along the whole circuit almost instantly, so every electron everywhere starts to drift at essentially the same moment — including those already in the lamp. Nothing has to travel from the cell to the lamp first.

A straw already full of peas: push one in, one drops out of the far end at once — and no single pea travelled the straw.